T.I.P. TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES

COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 - Data Structures and Algorithms | **SECTION:** | CPE21S3 |
| **DATE PERFORMED:** | 08/25/2026 | **DATE SUBMITTED:** | 08/25/2026 |
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**PRELIM EXAM** **PRELIM LABORATORY EXAM SET C**

## 1. PROCEDURE OUTPUT

**CODE:**

main.cpp > ...

```cpp
#include <iostream>
#include <string>

#include "queueLL.h"

bool isPalindrome(const std::string& number)
{
    Stack stack;
    Queue queue;

    for (char digit : number)
    {
        stack.push(digit);
        queue.enqueue(digit);
    }

    while (!stack.isEmpty())
    {
        char stackValue = stack.pop();
        char queueValue = queue.dequeue();

        if (stackValue != queueValue)
        {
            return false;
        }
    }

    return true;
}

int main()
{
    std::string number;
    std::cout << "Palindrome Number Checker:\n";
    std::cout << "Enter a integer: ";
    std::cin >> number;

    if (isPalindrome(number))
    {
        std::cout << number << " is a palindrome number.\n";
    }
    else
    {
        std::cout << number << " is not a palindrome number.\n";
    }

    return 0;
}

```

```cpp

#ifndef QUEUELL_H
#define QUEUELL_H

struct Node
{
    char data;
    Node* next;

    Node(char value)
    {
        data = value;
        next = nullptr;
    }
};

class Queue
{
private:
    Node* front;
    Node* back;

public:
    Queue()
    {
        front = nullptr;
        back = nullptr;
    }

    ~Queue()
    {
        while (!isEmpty())
        {
            dequeue();
        }
    }

    bool isEmpty() const
    {
        return front == nullptr;
    }

    void enqueue(char value)
    {
        Node* newNode = new Node(value);
        if (isEmpty())
        {
            front = newNode;
            back = newNode;
        }
        else
        {
            back->next = newNode;
            back = newNode;
        }
```

```cpp
#ifndef QUEUELL_H
class Queue
    Queue()
        front = nullptr;
        back = nullptr;
    }

    ~Queue()
    {
        while (!isEmpty())
        {
            dequeue();
        }
    }

    bool isEmpty() const
    {
        return front == nullptr;
    }

    void enqueue(char value)
    {
        Node* newNode = new Node(value);
        if (isEmpty())
        {
            front = newNode;
            back = newNode;
        }
        else
        {
            back->next = newNode;
            back = newNode;
        }
    }

    char dequeue()
    {
        if (isEmpty())
        {
            return '\0';
        }
        Node* temp = front;
        char value = temp->data;
        front = front->next;

        if (front == nullptr)
        {
            back = nullptr;
        }
        delete temp;

        return value;
    }
};
```

**OUTPUT**

| Palindrome? | Output (Console) | Remark |
|---|---|---|
| 10201 | | |
| 1003003001 | | |
| 88 | | |
| 0 | | |
| 202 | | |

**ANALYSIS**

**I just copy some code in my last activity, sorry sir i can't do this :((**

**PSEUDO CODE:**

**Start**
**Print " Palindrome Number Checker"**
**Print " Enter an integer:"**
**Input number**

**If is Palindrome (number**
**Print " is a palindrome number."**

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

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